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PV-ON: An online/bilingual PV sizing tool for grid-connected system, case studies in Egypt

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Summary

Photovoltaic (PV) grid-connected systems have shown a booming trend as a sustainable, green and low cost alternative for providing energy all over the world. Accordingly, PV sizing tools have been promoted to evaluate the expected performance and feasibility of the PV system at a given location. Various commercial as well as free/open-source PV sizing tool are currently available; however, most of them consider a single prospective, either technical or economic, without considering the location-based parameters such as the electrical tariffs. The aim of this article is to develop an online free PV pre-sizing tool for grid-connected system integrated with tariffs and electricity bills. The proposed tool can be accessed over the internet using multi-platforms (online), without any charges or subscription (free) and it generates a preliminary techno-economic solution showed in an online interface and downloadable PDF report (pre-sizing). Seeking for validation, various 24 case studies have been conducted over four different cities in Egypt, where a commercial tool has been utilized as a bench mark. An overall mismatching error various in the range of 1% to 6% has been recorded, showing the full appropriateness of the developed tool. Finally, a level of satisfaction questionnaire has been operated among various user segments, where acceptable outcomes have been observed.

KEYWORDS

electricity consumption, feasibility study, grid-tied, online simulations, PV sizing tools, solar energy

1 | INTRODUCTION

Photovoltaic (PV) is very promising source of energy if it is utilized accurately.¹ PV sizing tools are considered as one of the most widely techniques used to evaluate the potential of PV system at a given location. However, inappropriate

List of Symbols and Abbreviations: ξ_s , overall system efficiency; AMC, average monthly consumption; AMEB, average monthly electricity bills; AMGF, average monthly generation factor; APE, annual produced energy; AVA, available area; ASCD, annual saved carbon dioxide; CF, capacity factor; CPK, cost per kWp; CPK-AC, the cost per kW-AC estimated for inverters; INVC, the inverter cost; LCOE, levelized cost of energy; LCOE-Current, current levelized cost of energy for Egyptian Tariff; LCOE-PPA, levelized cost of energy for power purchase agreement option; LPE, lifetime produced energy; MC, maintenance cost; MGF, monthly generation factor; MP, maximum power; MSA, mounting system area; NOM, number of PV modules; OC, operation cost; P, peak power; P_{inv}, power of inverter; PP, The percentage power; Ps, selected power; RC, replacement cost; SEB, summer electricity bills; SFEB, spring and fall electricity bills; SPB, simple payback; SY, specific yield; Tariff, Egyptian Electrical Tariff; TP, total price; WEB, winter electricity bills.

sizing influences the investment and will lead to under/over sizing of the system. Consequently, pre-sizing accurately is the key parameter to guide solar installers, developers and non-technical users to implement PV systems effectively as well as to develop the continuation to PV industrial development.² The utilization of PV grid-connected system has proven to be a sufficient energy supply at high irradiation locations.³ Additionally, recent publications have shown the utility of such PV system with either mechanical tracking,⁴ or integrating artificial intelligent (AI) through maximum power point tracking optimizers.⁵

Several studies focused on technical sizing tools to study the technical analysis of grid-connected PV system were demonstrated in literature. Laib et al⁶ presented a simulation study on grid-connected system using Matlab-Simulink in order to study the performance and energy balance for residential house in Algeria. In Reference 7, the authors presented a dynamic model for grid connected system after using different approaches to fully utilize the PV system. The study concluded that the model was able to predict the shadow precisely. Shukla et al⁸ preformed a technical analysis for 110 kWp grid-connected system using Solargis PV planner software. The study shows that Solargis is an accurate, reliable and easy sizing tool which has the capability to model PV systems. In Reference 9, the authors conducted a study for 6.4 kWp grid-connected system using four types of sizing tools (SolarGIS, PVGIS, SISIFO and PV*SOL). The study concluded that PV*SOL showed to be reliable, easy and fast sizing tool with respect to others. A 100 MWp grid-connected PV system was carried out to study the technical analysis using PVsyst and SolarGIS tools.¹⁰ The study showed the ability of PVsyst to predict the yields accurately due to using detailed hourly data, moreover the capability of evaluating the PV system by importing the meteo-data at any location. Despite providing detailed technical sizing information, most of the above-mentioned tools are commercial and not capable of estimating the economic side of the PV project tested. Additionally, none of them utilize the tariff in the database which allows the user to size the PV system based on electricity bills.

On the other hand, some studies focused on economic sizing tools to study the economic analysis of grid-connected PV system. Mohammadi et al¹¹ discussed an economic study of 5 MWp grid-connected system using RETScreen tool and applied it to eight locations in southern coast of Iran. The study showed the capability of RETScreen for analysing the PV system. Duman and Güler¹² presented an economic study for grid connected system using HOMER tool and applied it in residential rooftop building in Turkey. The study demonstrated that HOMER enrolls 20 000 tariff (grid purchased) which is used in economic analysis and future projections. However, these tariffs are only for Canada, USA, and Mexico.¹² In Reference 13, the authors conducted an economic feasibility study using RETScreen tool for 5 kWp, the study was applied for 13 countries. They mentioned that there are some main factors affecting the viability of the system, which are electricity tariffs, solar irradiation, subsidies, and incentives. Ryan and La Monaca¹⁴ carried out an economic analysis for PV system rooftop in Ireland using System Advisory Model (SAM) tool which showed the capability to provide a financial modeling. Furthermore, they developed a methodology to reduce the years of payback while the study concluded that users with higher consumption have the chance to reduce the payback period of system. Feasibility study in England¹⁵ for 67 kWp grid-connected system was developed and tested. The authors created an economic analysis model and found that the payback for the system was 11 years. Despite the studies presented above, the technical analysis was mostly neglected whereas there should be a full techno-economic solution for sizing the PV system sufficiently.

Consequently, to study the techno-economic analysis of grid-connected PV system other studies used two types of sizing tools.^{4,16-18} Obeng et al¹⁹ performed a techno-economic feasibility study for 50 MWp grid-connected system at UENR Nsoatre campus, Ghana. The technical analysis was evaluated by PVsyst whereas economic analysis was estimated by RETScreen. An offline techno-economic tool was developed by using Excel combined by Microsoft visual in Malaysia.²⁰ The tool was found to be friendly interface as well as provided actual results by virtue of validation conducted to existing building. Nacer et al²¹ conducted a techno-economic study for grid-connected PV system in order to supply the electrical needs for a farm in Algeria. The study did not use any sizing tools for techno-economic evaluation, although calculations were conducted manually. In Reference 22, the authors illustrated a techno-economic study for three homes in Palestine. The technical analysis was carried out by PVsyst tool while economic analysis was applied by Excel software. Recently, the work in Reference 18 introduces a complete techno-economic study of a PV-grid connected system with DC batteries attached.

It can be concluded that all the above mentioned studies technical, economical, and techno-economic were applying tools to evaluate the PV grid-connected system. Hence, some studies were focusing on tools used for technical analysis others for economic analysis. Whereas the techno-economic studies were done by two types of tools one for technical other for economic analysis. Additionally, most of sizing tools presented are commercial and targeting technical users, leading that there are still some limitations to fulfill the simplicity of friendly interface for non-technical users. Equally

important, there is a lack of covering our local needs in Egypt by having a techno-economic PV sizing tool integrated with electrical tariffs to generate a full solution. In the same time, Middle East is rapidly improving in solar PV technology and there was not found a free trustable grid-connected sizing tool in Arabic language.

Based on literature review discussed, this article aims to fill the gaps by developing free online bilingual techno-economic PV pre-sizing tool for grid-connected system, taking into consideration simplicity and less running time. Herein, our proposed tool can be accessed through any online platform, with full free privileges, while demonstrating preliminary (pre-sized) output data that can be easily captured by technical as well as non-technical users. The proposed tool is integrated with electrical tariffs which will assist research-oriented users, developers and PV solar companies in preliminary design stage preventing them from over/under sizing the system. Moreover, the developed tool generates a full techno-economic solution presented in an online interface and downloadable PDF report. Herein, Egypt was selected as the location for implementing case studies in this tool due to its location in the Sunbelt as well as the business of PV system is booming in Egyptian markets nowadays. However, the expandability of the tool to include more locations is considered in the next version. The novel demonstrated tool is counted as the first online PV sizing tool with a complete Arabic version.

2 | MATHEMATICAL MODELLING AND GUI DEPLOYMENT

This section presents the methodology of modeling and deployment the proposed online pre-sizing tool (PV-ON). The tool is structured into three phases which started by presenting the inputs needed to size the system (cf. Figure 1). In the second and third phase technical and economical outputs are calculated based on the mathematical model introduced in this section. The demonstrated GUI in Figure 1 illustrates the overall view of the deployed PV sizing tool. The tool is launched via java scripting and deployed over the CELT domain. The coming paragraphs illustrate the overall process starting from the system inputs till the final report extraction (see Figure 1).

Initially, users are requested to input the data demonstrated in Figure 1 (steps 1 to 3). Users are asked to enter the type of system voltage, sector, location and available area. As a second step, user should indicate the type of the available area. Then, the user will be able to insert the monthly electricity bills for the four various seasons, assuming a constant load profile across each season. Following the input data given, the average monthly electricity bills (AMEB) can be calculated by:

USER INPUTS

Project's Info

1. System Voltage*
2. Sector *
3. Location *
4. Available Area [m²]

Type of Area



Ground Mounted



Rooftop



Industrial Shed

Electricity Bills

Please Insert one bill for the following Months		
No.	Select Month	Cost (EGP/Month)
1	December	Insert one month bill...
2	March	Insert one month bill...
3	June	Insert one month bill...

STEP1

STEP2

STEP3

* Drop-down menu

SYSTEM OUTPUTS

Technical Analysis

Example for Technical Outputs

Average Monthly Consumption	35,632.19	kWh
Power Required For covering the Consumption	207.364	kWp
Maximum Power based on the available area	199	kWp

Economic Analysis

Example for Economic Outputs

Cost Per kWp	13,000	EGP
Total Price	2,587,000	EGP
Simple Payback	4.35	Years

FIGURE 1 Graphical user interface (GUI) for the online PV sizing tool (PV-ON)

$$\text{AMEB} = (\text{WEB} + \text{SFEB} + \text{SEB})/3 \text{ (EGP/Month)} \quad (1)$$

where WEB is the winter electricity bills, SFEB is the spring and fall electricity bills, and SEP is the summer electricity bills. As result, the average monthly consumption for the four seasons will be calculated automatically and divided by the Egyptian tariff. Hence, the average monthly consumption (AMC) per month obtained by:

$$\text{AMC} = \text{AMEB}/\text{Tariff} \text{ (kWh/Month)} \quad (2)$$

where Tariff is the electrical Egyptian tariff in (EGP/kWh) depends on the type of system voltage and sector, more information is provided in Appendix A.

Two different types of DC electric power will be then calculated. The first is the peak power (P) which is calculated based on the electricity bills inserted. It is determined by dividing AMC over the average monthly generation factor²³:

$$P = \frac{\text{AMC} \text{ (kWh/Month)}}{\text{AMGF} \text{ (kWh/kWp/Month)}} \text{ (kWp)} \quad (3)$$

where AMGF is the average monthly generation factor obtained by:

$$\text{AMGF} = [\text{The average monthly irradiation kWh/m}^2/\text{year}] \times [\xi s] \quad (4)$$

where ξs is defined as the overall system efficiency which equals to 78%. Hence, the total detailed losses are demonstrated in Appendix B.

Second, the maximum power (MP) is calculated to maximize the installed power based on the inserted available area as well as its selected mounting type, MP is defined by:

$$\text{MP} = \frac{\text{AVA}}{\text{MSA}} \text{ (kWp)} \quad (5)$$

where AVA is the inserted available area (m^2), while MSA is the mounting system area (in m^2/kWp), and its defined as the total area needed per kWp installed with respect to the chosen area type. The value of MSA is assumed, based on industrial imperial calculations, to be $10 \text{ m}^2/\text{kWp}$ for rooftop, $12 \text{ m}^2/\text{kWp}$ for ground mounted and $8 \text{ m}^2/\text{kWp}$ for ground mounted.

The percentage power (PP) is obtained by dividing the maximum power (MP) by peak power (P)

$$\text{PP} = \left(\frac{\text{MP}}{P} \right) * 100 (\%) \quad (6)$$

It is calculated to evaluate the percentage of covering the consumption in order to avoid under/over sizing. As well as to guide the user which power will be selected in sizing. Accordingly, if the $\text{PP} \geq 100\%$, the system will be sized based on the calculated peak power (P), conversely if $\text{PP} < 100\%$ the system will be sized based maximum power (MP).

After selecting which power will be used in sizing, the AC power generated from inverter (Pinv) is obtained by dividing it by DC/AC ratio. Hence, DC/AC ratio ranges from 10% to 20% less than the DC power. Herein, it is assumed to be 20%. Accordingly, (Pinv) is calculated by:

$$\text{Pinv} = \frac{P_s}{1.2} \text{ (kW - AC)} \quad (7)$$

where P_s is the DC selected power based on Equation (6).

The number of PV modules (NOM) is calculated by dividing the selected power (P_s) over the power of PV module which is obtained by:

$$\text{NOM} = \text{Ps}/0.335 \quad (8)$$

here the selected module's power and efficiency are found to be 335 W and 17.01%, respectively, considering a commercial PV module in Reference 24.

Annual produced energy (APE) across the year depends on which type of selected area, if the area is rooftop or ground mounted system, it is calculated by:

$$\text{APE} = \text{Ps} * \text{MGF} \text{ (kWh/year)} \quad (9)$$

where MGF is the monthly generation factor obtained by:

$$\text{MGF} = [\text{The monthly irradiation (kWh/m}^2\text{/month)}] \times [\xi_s] \quad (10)$$

Moreover, if the selected area is industrial shed, a loss of 5.5% is applied due to the change from optimum angle to industrial shed which is obtained by:

$$\text{APE} = (\text{Ps} * \text{MGF}) * 0.945 \text{ (kWh/year)} \quad (11)$$

The generated energy produced per kWp installed across the year is defined as the specific yield (SY) which can be obtained by:

$$\text{SY} = \text{APE}/\text{Ps} \text{ (kWh/kWp/Year)} \quad (12)$$

Additionally, the capacity factor (CF) is determined based on the location selected, the better the localized PV system the higher the capacity factor the more energy captured in daytime for a given PV system components,²⁵ it can be determined by:

$$\text{CF} = \text{SY}/\text{Operating Time (\%)} \quad (13)$$

where the Operating time is the total hours across the year 8760 hours. It is also defined as the maximum energy that can be generated at full rated power for 24 hours/day.²⁶

Following the PV module specs presented in,²⁴ the lifetime of the PV module is shown to be 25 years. Therefore, the lifetime produced energy (LPE) from year 1 to year 25 is calculated by summation all APE, taking into consideration a reduction of 2.5% as percentage of degradation for the first year and 0.05% for the remaining years:

$$\text{LPE} = \sum_{1}^{25} \text{APE} \text{ (kWh)} \quad (14)$$

The avoided annual saved carbon dioxide (ASCD) which is generated from conventional power plants and other sources of energy is obtained by:

$$\text{ASCD} = (\text{APE}) * \text{Factor for CO}_2 \text{ Avoidance} \quad (15)$$

where Factor for CO₂ Avoidance is taken to be 0.6 kg/kWh.²⁷

On the other side, an economic feasibility for the system is obtained through calculating a set of economic parameters. The price per kWp (CPK) changes according to the installed capacity of the PV system. The prices of installed power ranges from 5 kWp to 60 MWp which cover EPC (Engineering, Procurement, and Construction), including price covers PV panels, inverters, balance of the system and construction, more details can be addressed in Appendix C1. Subsequently, the total price (TP) of the PV system will be calculated by:

$$TP = CPK * Ps \text{ (EGP)} \quad (16)$$

The simple payback (SPB) shows the number of years which the user will recover his investment during PV system lifetime which can be obtained by:

$$SPB = TP / (APE * \text{Tariff (Years)}) \quad (17)$$

The levelized cost of electricity (LCOE), which is the cost per kWh for the PV system, is presented in the tool in three alternatives. The first is the current Egyptian tariff used from government (LCOE-Current). Additionally, power purchase agreement (LCOE-PPA) where the investor will supply the user energy through 25 years, usually percentage ranges from 15% to 20% less than the current Egyptian tariff, herein is assumed to be 15%. Furthermore, the third option is by owning PV system which covers full turnkey price for engineering, procurement and construction (LCOE-EPC) including lifetime operation cost for 25 years:

$$LCOE = \frac{TP + MC}{LPE} \text{ (EGP/kWh)} \quad (18)$$

where MC is the maintenance cost which covers operation (checking and cleaning) and inverter replacements over 25 years. The maintenance cost can be defined by:

$$MC = OC + RC \text{ (EGP)} \quad (19)$$

where OC is the operation cost which covers the system operative and predicative maintenance including cleaning, a full detailed percentages and costs over the lifetime of the PV system shown in Appendix C2. RC is replacement cost of inverter (Prices per kW-AC for all capacities is provided in Appendix C3. Hence, (RC) is changed two times only (during the eighth and 16th years of operation), calculated by:

$$RC = INVC * ((1 + 0.11)^8 + (1 + 0.11)^{16}) \text{ (EGP)} \quad (20)$$

where INVC is the present inverter cost, assuming an inflation rate of 11%.

$$INVC = P_{inv} * CPK - AC \text{ (EGP)} \quad (21)$$

where CPK-AC is the cost per kW-AC estimated for inverters. In this context, we are considering a string inverter for the small and medium scale, as well as central inverters for mega scale projects. Whereas the maximum efficiency for the inverter is 98.9%.²⁸

Based on the mathematical procedure demonstrated in this section, an offline as well as an online pre-sizing tool have been deployed. The offline version has been scripted using Matlab, available online through.²⁹ The code is also attached in the Supplementary Material S-1. The online version has been scripted using Java-script and deployed in Reference 30. In addition, the tool can be operated offline as a mobile app through Reference 31. A detailed step by step guiding video is also included in the supplementary material S-2 and can be accessed via Reference 32.

3 | CASE STUDIES AND VALIDATION

In this section, 24 case studies were sized using the proposed tool, classified into three categories (mega, medium and small scale) based on the case study peak power (cf. Figure 2). This large variation in the system peak power from 5 kWp to 2 MWp has been investigated to check the validity of our proposed pre-sizing tool with respect to other commercial tools. This will be highlighted in more details by the end of this section. Additionally, the variety was applied by using different types of aspects in order ensure the ability of covering the needs of technical, non-technical user and developers. As a result, the variety in case studies includes:

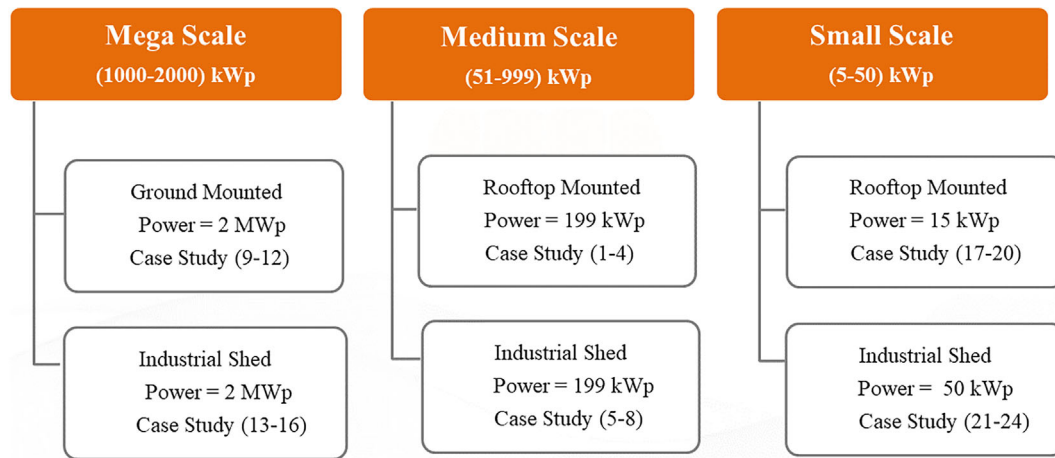


FIGURE 2 Conducted 24 case studies

STEP 1

Insert Project's Name
Case Study 3

Select System Voltage
Low

Select Your Sector
Residential

Select Your Location
Aswan

Insert Your Available Area in (m²)
1990

STEP 2

Select Type of Area

Ground Mounted Rooftop Industrial Shed

STEP 3

Please Insert one bill for the following Months

No.	Select Month	Cost (EGP/Month)
1	December	40000
2	March	50000
3	June	65000

FIGURE 3 Implemented inputs for case study 3

- Cities implemented in the tool (Cairo, Alexandria, Aswan and Hurgada).
- System Voltage (Medium /Low).
- Sector Type.
- Different types of available Area.
- Variation in Electricity Bills.

Furthermore, case study 3 will be presented in detailed manner while the remaining 23 case studies can be found in (Supplementary Material S-3).

The inputs implemented in the Online PV Sizing tool for case study 3 are shown in main three steps adjusted according to the user needs as shown below in Figure 3. The system outputs are generated in a form of two sets of results; technical and economic outputs (cf. Figures 4 and 5). Additionally, the complete case study can be also shown in a form of downloadable PDF report as indicated in Figure 5, where save as PDF button can be clicked. Finally, in order to capture our targeted user feedback and comments, a survey questionnaire is embedded as a final stage in the tool where users can evaluate the tool technically and graphically as well as write comments (see Figure 6).

Technical Analysis					
Average Monthly Consumption	35,632.19	kWh	Annual Produced Energy	410,338.04	kWh
Power Required For covering the Consumption	207.364	kWp	Annual Specific Yield	2,062	kWh/kWp
Maximum Power based on the available area	199	kWp	Life Time Produced Energy (25 Years)	9,479,861.43	kWh
Percentage of covering the Consumption	95.97	%	Circuit Breaker Capacity	400	A
Number Of Modules Using Jinko - 335 W	594	Units	Annual Saved Carbon Dioxide	246,202.82	kg
Power Of Inverter	165.83	kW-AC	Tilt Angle of the PV Panels	25	Degree

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Next >

FIGURE 4 Technical outputs for Case Study 3

Economic Analysis					
Cost Per kWp	13,000	EGP	Current Cost of kWh LCOE – Current	1.45	EGP/kWh
Total Price	2,587,000	EGP	Total Saving Money	13,745,799.07	EGP
Simple Payback	4.35	Years	Cost of kWh (Owning PV System)	0.696	EGP/kWh
Cost of kWh Power Purchase Agreement (PPA)	1.23	EGP/kWh			

*PPA is a contract between two parties, one which generates electricity and one which is looking to purchase electricity.

*LCOE is Levelized Cost of Energy (EGP/kWh).

< Back
Finish
Save Data as PDF

FIGURE 5 Economic outputs for Case Study 3

Seeking for a concrete validation, the results for case study 3 are validated by PVsyst software. A perfect matching was observed in Table 1 with a total percentage of error of 1%. Consequently, the other 23 case studies conducted have resulted a percentage of error ranges from 1% to 6% (the complete validation process for another 7 cases is demonstrated in supplementary material S-5). It can be noted that the variation in percentage of error is due to the irradiation database, where PVsyst has irradiation data from Metronome 7.2 and the irradiation data used in our proposed online

1.How would you rate our PV Sizing Tool?

Design of the site.	★ ★ ★ ★ ★
Technical Descriptions	★ ★ ★ ★ ★
Economical Descriptions.	★ ★ ★ ★ ★
Finding what you needed.	★ ★ ★ ★ ★

2.Thanks for that, now could you tell us a bit about the simulation?

How would you rate the time of Simulation?	★ ★ ★ ★ ★
The friendliness of website interface?	★ ★ ★ ★ ★
The results of the simulation you received.	★ ★ ★ ★ ★

3.Could you tell us from where you Sized your PV system?

☒ Smartphone
 ☐ Laptop/Desktop
 ☐ Laptop/Desktop

4.Any Suggestions For Us?

Write Your Suggestions...

Finally, Did you like the idea of Sizing Your system by your ownself ?

★ ★ ★ ★ ★

Submit Survey

FIGURE 6 Online survey form

TABLE 1 PVsyst Validation results for Case study 3

Technical comparison	Online PV sizing tool	PVsyst software
Power (kWp)	199	199
Annual produced energy (MWh)	408.07	412.70
Specific yield [(kWh/kWp)/year]	2050.61	2073.87
Capacity factor (%)	23.4	23.7
Percentage of error (%)	1	

PV pre-sizing tool is from PVGIS software. However, the percentage of error is acceptable whereas a conducted study for 1 MWp using PVGIS and PVSyst in Reference 25, where the authors concluded that there was a deviation around 8%-15% in output results between the two sizing tools.

4 | USERS EVALUATIONS AND FEEDBACK

In order to evaluate the proposed tool from our targeted users prospective, a beta version of the tool has been published using social media platforms. Our targeted users where selected to be either from the business sector, technical users from academia, postgraduate as well as senior undergraduate students and non-technical users. As mentioned earlier, a survey was associated to the tool to gather the user's feedback and comments. The statistics showed that, there were 46 case studies conducted by the users. Table 2 shows the number of simulations for both medium and low voltages (complete details is found in supplementary material S-6). It can be noted that the low voltage has the largest number of simulations as this voltage covers different sectors which covers residential, commercial, irrigation, general lighting and other users. Moreover, for medium voltage the other users used in sizing was the industrial users which covers were mainly the factories. One of the inputs needed by the user is the sector type, herein a distribution of the selected sector type is indicated in Figure 7A. Residential sector recorded the highest percentage of 66%, where this is attributed to the present trend in Egypt associated with new cities construction. The distributions of the simulations trials made by users over the cities are illustrated in Table 3 as well as Figure 7B. It should be pointed out that Cairo has the highest number of simulations, this is due to the development of implementing the grid-connected PV system in the new buildings.

Following the conducted survey, Table 4 shows the average survey results taken from 22 users. The results showed a satisfaction from users, where this tool is helpful and friendly interface. However, users suggested that the results of simulations can be shown better by improving some futures like icons and graphics.

TABLE 2 Number of Simulation conducted by users for low and medium voltage

System voltage	Sector	Number of simulation	Total number of simulations
Low	Residential	25	38
	Commercial	4	
	Irrigation	2	
	General lighting	6	
	Other users	1	
Medium voltage	Water and sanitation company	0	8
	Irrigation	0	
	Other users	8	
Total number of simulations			46

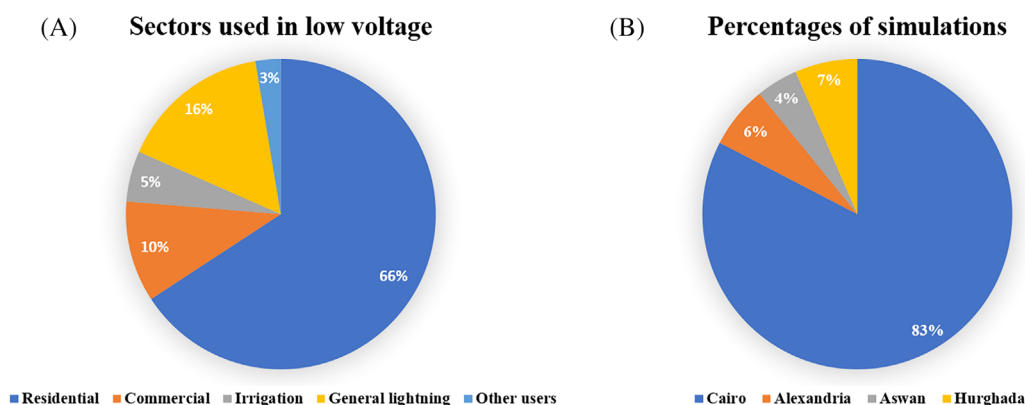


FIGURE 7 Percentages of using sectors in low voltage (A) and cities (B)

TABLE 3 Statistical analysis for the cities used in sizing

City	Total number of simulations
Cairo	38
Alexandria	3
Aswan	2
Hurghada	3

TABLE 4 Average ratings from users survey feedback

Term	Rating (1–5)
Design of the site.	4.5
Technical descriptions	4.4
Economical descriptions	4.3
Finding what user needed	4.2
Time of simulation	4.5
Friendliness of website interface	4.4
Results of the simulation	4.2

5 | CONCLUSION

The developed online PV pre-sizing tool has been presented including the capabilities of covering the gaps between the current PV sizing tools nowadays. The method used shows the simplicity of interface in order to be applicable for developer, technical and research-oriented users. Moreover, the integration of tariff database for both medium and low voltage showed the capability to size the PV system based on available electricity bills, as well as the ability of sizing from 5 kWp to 2 MWp which covers our local needs. Additionally, 24 case studies for mega, medium and small scales were conducted and validated by PVsyst sizing tool, correspondingly the results showed a percentage of error ranges from 1% to 6%. Moreover, a statistics results were discussed for the user who used the sizing tool and conducted the survey. It can be noted that the developed tool is the first free online grid-connected pre-sizing tool in Arabic version which generates a techno-economic solution shown in the interface or can be downloadable PDF file and it can be applied at any country (Arabic version is presented in supplementary material S-7). The developed tool can be improved by implementing a database for inverters and PV modules which make it adjustable to technical users. However, financial modeling feature may be more attractive for investors.

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PEER REVIEW

The peer review history for this article is available at <https://publons.com/publon/10.1002/2050-7038.12910>.

DATA AVAILABILITY STATEMENT

The data that support the findings of this study are available as following^{29–33}:

The offline version has been scripted using Matlab, available online through:

<https://www.mathworks.com/matlabcentral/fileexchange/75738-pv-on-offline-pv-grid-connect-system-pre-sizing-tool>

The code is also attached in the Supplementary Material S-1. The online version has been scripted using Java-script and deployed through:

<https://www.celt-bue.com/#!/restricted/viewOnlineToolPVsizing>

In addition, the tool can be operated offline as a mobile app through: <https://celt-bue.netlify.app/#/>. A detailed step by step guiding video is also included in the supplementary material S-2 and can be accessed through:

<https://www.youtube.com/watch?v=M4kQHnSSCzk&t=4s>

Egyptian governmental tariffs can be accessed through:

<http://egyptera.org/ar/Tariff2019.aspx>

Any other data that support the findings of this study are available from the corresponding author upon reasonable request.

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SUPPORTING INFORMATION

Additional supporting information may be found online in the Supporting Information section at the end of this article.

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APPENDIX A: Egyptian electricity tariffs

This section demonstrates the Egyptian electricity tariffs for both low voltage (cf. Table A1) and medium voltage (cf. Table A2) consumptions among different sectors.

TABLE A1 Low voltage tariff

Sector	Average monthly electricity bills (AMEB)	Tariff (EGP/kWh)
Residential	AMEB Less than 650EGP	1
	AMEB are $\leq$ 1000 EGP	1.40
	AMEB are >1000 EGP	1.45
Commercial	AMEB are Less than 840 EGP	1.40
	AMEB $\leq$ 1550 EGP	1.55
	AMEB >1550 EGP	1.60
Irrigation ^a	—	0.75
Water and sanitation companies ^a	—	1.25
Other users ^a	—	1.25
General lighting ^a	—	1.25

^aThe cost of tariff is constant for all consumption levels.

TABLE A2 Medium voltage tariff

Sector	Average monthly electricity bills (AMEB)	Tariff (EGP/kWh)
Irrigation purposes ^a	—	0.99
Water and sanitation companies ^a	—	1.20
Other users ^a	—	1.15

^aThe cost of tariff is constant for all consumption levels.

APPENDIX B: System losses and efficiency breakdown

See Table B1.

TABLE B1 Applied losses

No	Losses	Percentage losses (%)	System efficiency (%)
1	Total sunlight reaching the PV Modules losses	6.55	0.936
1.1	IAM factor on global	2.55	0.975
1.2	Soiling loss factor	3	0.960
2	Total DC power generation from PV modules losses	14.5	0.855
2.1	PV loss due to irradiance level	0.42	0.996
2.2	PV loss due to temperature	8.8	0.912
2.3	Module quality loss	0.74	0.993
2.4	Light-induced degradation	2	0.980
2.5	Module array mismatch loss	1.14	0.989
2.6	Ohmic wiring loss	1.4	0.986
3	Total DC to AC electricity conversion Losses	2.48	0.975
3.1	Inverter loss during operation	1.1	0.984
3.2	System unavailability or loss due to available factory	0.86	0.991
3.4	Night consumption	0.01	1.000
3.5	Inverter loss due to power threshold	0.01	1.000
Total losses		22.03	78

APPENDIX C: PV system cost breakdown

Ranges of power (kWp)	Cost per kWp (EGP)	Cost per kWp (USD)
5-50	14 000	875
51-100	13 500	844
101-500	13 000	813
501-1000	10 610	663
1001-5000	10 200	638
5001-60 000	10 000	625

TABLE C1 Cost per kWp (CPK-DC)

C1 | PV system cost per kWp

Table C1 illustrates the needed cost per kWp for a PV system including Engineering, Procurement, and Construction (EPC) expanses in the term of PV panels, inverters, and balance of the system and construction costs. Herein, a detailed breakdown for the initial investment of presented case study 3 (199 kWp) is illustrated. Whereas, the prices of PV panels, inverters, balance of the system, constructions are 5.7 EGP/W, 1.72 EGP/W, 0.51 EGP/W, and 1.4 EGP/W, respectively. As result, these prices are represented as percentages in Figure C1. Knowing that, the maximum efficiency for the selected inverter is 98.9%. Additionally, a detailed pricing sheet for 199 kWp is provided in supplementary material S-8.

C2 | Operation and cleaning cost

See Table C2.

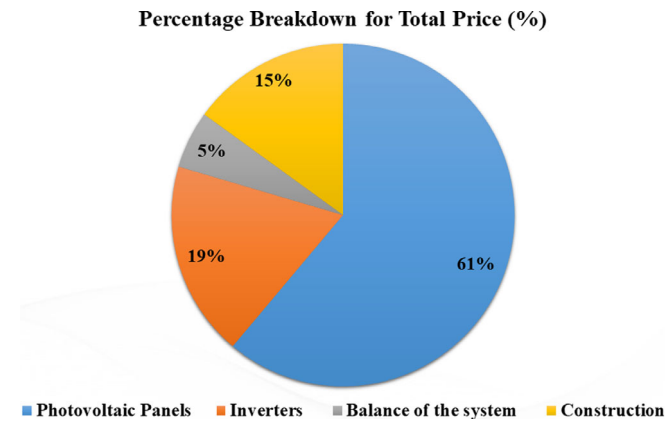


FIGURE C1 Cost breakdown for case study 3

TABLE C2 Operation and cleaning cost

Year	Price of the operation cost including cleaning
1	No charges
2–5	3% from the total cost
5–10	4% from the total cost
10–20	5% from the total cost
10–20	7% from the total cost
Total	1.20% From The total Cost

C3 | Inverter replacement cost

The cost per kW-AC changes according to the capacity of the system. Herein, Table C3 illustrates the estimated cost per kW-AC using string and central inverters. Consequently, this cost has been integrated in cash-flow after including the inflation rate (11%) for the eighth and 16th years as described in Equation (20).

TABLE C3 Inverter replacement cost

Power of inverter (P _{inv}) kW-AC	Cost per kW-AC EGP/kW-AC
P _{inv} = 5	5886
P _{inv} > 5 & P _{inv} ≤ 15	3100
P _{inv} > 16 & P _{inv} ≤ 30	2420
P _{inv} > 31 & P _{inv} ≤ 60	1720
P _{inv} > 61 & P _{inv} ≤ 100	1230
P _{inv} > 101 & P _{inv} ≤ 500	716
P _{inv} > 501 & P _{inv} ≤ 1000	700
P _{inv} > 1001 & P _{inv} ≤ 5000	700
P _{inv} > 5001 & P _{inv} ≤ 60 000	650